

DETECT · PROTECT · EXPLAIN

# Detector

*Detect phishing. Protect users. Explain every score.*

A local-first, self-hosted phishing and suspicious-website detection web application. Paste any URL and Detector returns a 0–100 risk score — classified Safe, Suspicious, or Phishing — with fully explainable reasons behind every verdict.

● SECURITY ENGINEERING · PRODUCT PRESENTATION · 2026



# Agenda

## 01 Problem

The phishing epidemic & India-targeted scams

## 03 Solution

What Detector is & why it differs

## 05 How It Works

Pipeline, heuristics, enrichment

## 07 Proof & UX

Demo results, dashboard, exports

## 02 Gap

Why existing scanners fall short

## 04 Features

Detection, scoring & reporting

## 06 Scoring

3-layer fusion model & verdicts

## 08 Getting Started

Install, deploy, roadmap & Q&A



Detector · Detection Deck

## 01 - PROBLEM

# Phishing is exploding — and India is a prime target

Fake betting apps, gambling sites and brand-impersonation scams are flooding the region, and ordinary users cannot reliably tell a real site from a convincing fake.

## APAC CLICK RATE

## 5.5×

Asia-Pacific users click phishing links at nearly double the global rate per 1,000 users

## AI-GENERATED SHARE

## 80%

of detected phishing volume in APAC is AI-generated, with click rates exceeding 50%

## CYBERCRIME SHARE

## 30%

of all recorded crime in half of surveyed APAC countries is now cybercrime (INTERPOL)

SOURCES: INTERPOL APAC THREAT ASSESSMENT 2025/26 - THREATBOOK 2026 MID-YEAR APAC REPORT

## INDIA-MARKET THREAT

**Fake betting & gambling apps** and **brand impersonation of Indian banks and payment apps** are among the fastest-growing fraud vectors.

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Attackers blend convincing brand look-alikes with illegal content, making them hard for both automated scanners and humans to judge.





## 01 · THE GAP

# Why existing tools fall short

The four failure modes of today's scanners — each leaves a gap that Detector is built to close.

01

## Privacy leak

Free online scanners send your URL to third parties — your investigation becomes someone else's data.

02

## Black-box scores

Enterprise tools return an opaque number with no reasoning you can audit or trust.

03

## No help for humans

Ordinary users cannot tell a convincing fake from a real site — and free tools don't teach them.

04

## Costly AI dependency

AI-only detection charges per call, depends on the cloud, and explains nothing.



FREE = LEAKY · ENTERPRISE = OPAQUE · AI = COSTLY

# Meet Detector

A private, free, explainable alternative that anyone can self-host — built around a deterministic local heuristic engine, with optional threat feeds layered on top.



## Local-first

Core detection runs as a local rule engine — no paid AI dependency, no mandatory third-party data sharing.



## Explainable

Every score ships contributing & mitigating factors grouped into Impersonation, Domain Trust, Page & Content, and Infrastructure.



## Self-hosted

Runs on your machine or your server — students, analysts, small organizations and privacy-conscious users included.

### HOW IT WORKS IN ONE LINE

Paste a URL → Detector returns a **0–100 risk score** classified as **Safe (0–24)**, **Suspicious (25–49)** or **Phishing (50–100)** — with the reasons behind every verdict.

External threat feeds are **optional enrichments**; if no API keys are configured, enrichment is silently skipped and local analysis still runs fully.

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# What makes Detector different

Four design decisions that turn a scanner into a trustworthy, private, self-hostable security tool.



01

## Local-first & private

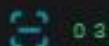
Deterministic local heuristic engine — no paid AI dependency, no mandatory third-party data sharing.



02

## Optional enrichment

VirusTotal, Safe Browsing, urlscan.io and AbuseIPDB activate only when you add your own free API keys.



03

## Explainability

Contributing & mitigating factors per verdict, grouped into Impersonation, Domain Trust, Page & Content, Infrastructure.



04

## India content-policy layer

Detects gambling, Indian betting apps and adult content; brand + illegal content is forced to a phishing verdict.



PRIVATE · FREE · EXPLAINABLE · INDIA-AWARE

# Detection & scoring features

The core detection stack — every stage contributes signal, every signal feeds the final score.



01

## Explainable risk scoring

0–100 score with contributing & mitigating factors for every verdict.



02

## 8-stage pipeline

Normalize → fetch → inspect → extract → enrich → classify → score.



03

## URL heuristics

Typosquatting, suspicious keywords, raw-IP URLs, risky TLDs, shorteners.



04

## Domain intelligence

WHOIS age buckets, SSL certificate checks, MX-record presence.



05

## Content-policy layer

Gambling, Indian betting and adult-content classification.

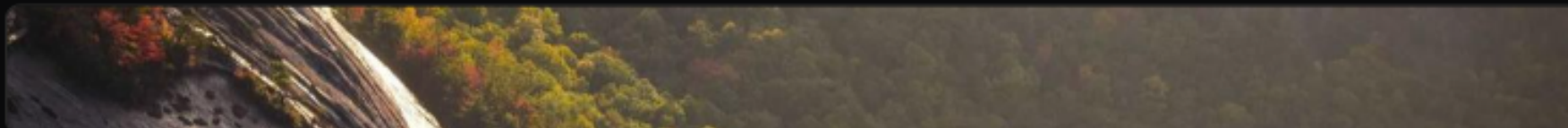


06

## Fused scoring

3-layer fusion with trust offsets and override floors, persisted to SQLite + JSON.

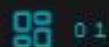
8 STAGES · 4 SIGNAL  
GROUPS · 1 FUSED SCORE





# Report, export & operate

From a rich dashboard to one-click exports — how Detector presents findings and stays configurable.



01

## Visual dashboard

Radial score gauge, score-composition bar, domain-age bar, redirect tracker, provider status cards.



02

## Two-tier PDF export

Standard Analysis Report + Extended Footprint Report with full threat-feed intelligence.



03

## JSON export

Machine-readable record of every analysis, ready for automation and archival.



04

## Installable PWA

Offline app shell, cached last-10 results, push notifications on risky verdicts.



05

## User feedback

One-click satisfied / not-satisfied capture, stored in the database for iteration.



06

## ~100 tunable knobs

Thresholds, keyword lists and brand lists configured via a single .env file.





# Built on a clean, layered stack

Five layers — from browser to storage — each with focused, well-trodden tools.

01 · FRONTEND

**Jinja2 templates · Vanilla JS · Chart.js**

Server-rendered pages, interactive charts, installable PWA shell (service worker + manifest)



02 · BACKEND

**Python 3.12 · Flask 3 · SQLAlchemy**

Lightweight application server and ORM with SQLite / Postgres support



03 · ANALYSIS

**BeautifulSoup · python-whois · dnspython · OpenSSL**

HTML parsing, WHOIS lookups, DNS checks and TLS certificate inspection



04 · STORAGE

**SQLite (default) · Postgres via DATABASE\_URL**

Persisted analyses, history, feedback and scores

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05 · SECURITY**Flask-Talisman · CSRF · Rate limit · SSRF guard**

CSP/HSTS headers, frame-DENY, 30/hour rate limit, structured JSON logging with rotation

09 / 22



# Architecture — the 8-stage detection pipeline

Every URL flows through eight stages; each stage feeds structured signals into the fused scorer.

01

## Normalize & validate

SSRF guard blocks private and loopback IPs

02

## Fetch page

Retries, 5-hop redirect cap, TLS checks, bot-block detection

03

## Inspect content

Password fields, external forms, iframes, ad scripts, favicon

04

## Extract URL features

Typosquatting, keywords, risky TLDs, shorteners, HTTPS

05

## WHOIS intelligence

Domain age buckets, SSL cert, MX records

06

## Enrich (optional)

VirusTotal, Safe Browsing, urlscan.io, AbuseIPDB

07

## Content policy

Gambling, Indian betting, adult classification

08

## Fused scoring

3-layer fusion, trust offsets, override floors → SQLite + JSON

8 STAGES · END-TO-END  
VERIFIED



# The visual report dashboard

One screen that turns a score into a story a non-technical viewer can read at a glance.



## Radial score gauge

Animated fill showing the 0–100 risk score and its band.



## Score-composition bar

Stacked bar showing how each layer contributed to the total.



## Domain-age threshold bar

Green / yellow / red zones mapping age to risk.



## VirusTotal charts

Engine flags and community votes visualized.



## Redirect-chain tracker

Step by step, what a URL takes before landing.



## Provider status cards



# Share findings in the format you need

Three export paths — from a human-readable summary to raw machine data.

## TIER 01

[PDF](#)

### Standard Analysis Report

A concise, professionally formatted summary built from report cards.

- Final verdict & score
- Contributing factors
- Key domain & content facts

## TIER 02

[PDF](#)

### Extended Footprint Report

Full threat-feed intelligence when enrichment is enabled.

- VirusTotal engine & votes
- Safe Browsing matches
- AbuseIPDB & urlscan.io data

## MACHINE

[JSON](#)

### JSON Export

A structured record of every analysis, ready for scripts and archival.

- Full signal breakdown
- Per-layer scores
- Timestamps & metadata





# Security by design

The tool that inspects untrusted URLs is hardened so inspecting them can't hurt you.



01

## Local-first privacy

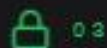
Core detection runs locally — no mandatory third-party data sharing.



02

## SSRF guard

Blocks requests to private and loopback IP addresses.



03

## Flask-Talisman

Strict CSP, HSTS and frame-DENY headers on every response.



04

## CSRF + rate limit

CSRF tokens on all forms and a 30-requests/hour rate limit.



**Operational hygiene:** structured JSON logging with rotation; optional threat feeds require your own keys; packaged EXE and launchers for trusted environments. HARDENED BY DEFAULT

# Install & run in six steps

No cloud account, no API key required to start — keys only add optional threat feeds.

1

## Clone the repository

Get the Detector source on your machine.

2

## Copy .env.example to .env

Optional: add free API keys for VirusTotal, Safe Browsing, AbuseIPDB, urlscan.io.

3

## Create a venv & install

Set up a virtual environment and install requirements.txt.

4

## Run the app

python run.py — or double-click start.bat / start.sh which auto-creates the venv.

5

## Open & analyze

Open `http://127.0.0.1:5000`, paste a URL, view the full report.

6

## Optional extras

Install as a PWA, or run the packaged single-file Detector.exe — no Python needed.

### VERIFIED DEMO

|         |            |
|---------|------------|
| URL     | google.com |
| Score   | 5 / 100    |
| Verdict | Safe       |
| Status  | Reachable  |



# Run it your way

From a dev laptop to a packaged desktop binary — choose the path that fits your environment.

## OPTION 01

[DEV](#)

### Source run

python run.py — or start.bat / start.sh which auto-creates the environment.

## OPTION 02

[DESKTOP](#)

### Packaged EXE

PyInstaller single-file Detector.exe (~57 MB) — no Python required.

## OPTION 03

[DATA](#)

### Database choice

SQLite by default; Postgres supported via the DATABASE\_URL variable.

## OPTION 04

[SCALE](#)

### Multi-worker

Redis-backed rate limiter supported for multi-process deployments.

SOURCE · EXE ·  
SQLITE/POSTGRES · REDIS

# Limitations & future scope

Transparency builds credibility — here's where Detector stands today and where it's headed.

## ⚠️ Current Limitations KNOWN GAPS

- **No automated test suite yet** — manual test cases only; CI needs dev tooling + a Dockerfile.
- **Slower analysis** — blocking WHOIS and VirusTotal lookups can take several seconds.
- **Production hardening** — move to gunicorn; enable secure cookies behind HTTPS.

## 🔗 Future Scope ROADMAP

- **ML models** as an additional detection layer on top of heuristics.
- **Browser extension** for in-browser URL checks.
- **Multi-user auth**, more threat feeds, and full CI/CD.

HONEST TODAY ·  
AMBITIOUS NEXT





## 08 · THE TAKEAWAY

# Detect phishing. Protect users. Explain every score.

A private, free, self-hosted detection tool built for students, analysts, small organizations and privacy-conscious individuals.

**Local-first**

Private by default

**Zero AI cost**

No paid API needed

**Transparent**

Every score explained

**India-aware**

Betting &amp; brand scams

## REFERENCES

VirusTotal · Google Safe Browsing · AbuseIPDB · urlscan.io — documentation & free-tier APIs

# Thank you.

[QUESTIONS & ANSWERS](#)